



# Pak-Austria Fachhochschule:

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## Minutes of the DGC meeting for MS Research Work Evaluation of Ms. Sabahat Siddique

Questions to clarify:

1- What is Cognitive Requirement Engineering, Examples?

Cognitive Requirement Engineering (CRE) extends traditional requirements engineering by integrating principles from cognitive science to design AI systems that align with human cognitive processes, mental models, and decision-making patterns. It focuses on how users perceive, understand, trust, and interact with AI systems.

**Examples:**

| Application Area       | Traditional RE Approach                                | Cognitive RE Approach                                                                                                                                                                                                                 |
|------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Medical Diagnosis AI   | "The system shall provide diagnosis with 95% accuracy" | "The system shall present diagnostic conclusions in a format that matches radiologists' mental models, highlighting anomalies in order of visual salience and providing confidence scores calibrated to clinical uncertainty metrics" |
| Autonomous Vehicle HMI | "The vehicle shall display system status"              | "The interface shall convey system intent and confidence levels using multimodal cues (visual, auditory, haptic) that align with driver expectations and minimize cognitive load during critical maneuvers"                           |
| Financial Advisory AI  | "The system shall recommend investment portfolios"     | "Recommendations shall be accompanied by explanations structured according to investors' mental accounting frameworks, with risk information presented using cognitive biases-aware visualization techniques"                         |

2- What are Ethical AI systems, Examples?

Ethical AI systems are artificial intelligence systems designed, developed, and deployed with intentional consideration of moral principles and societal values, ensuring they operate beneficially, justly, and transparently while minimizing potential harms.

| System Type             | Ethical Implementation                                              | Key Features                                                                                                                                                          |
|-------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IBM Watson for Oncology | Evidence-based treatment recommendations with citation transparency | <ul style="list-style-type: none"><li>• Provides supporting research for each recommendation</li><li>• Acknowledges confidence levels and evidence strength</li></ul> |

|                                      |                                         |                                                                                                                                                                                                          |
|--------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      |                                         | <ul style="list-style-type: none"> <li>• Designed for physician oversight, not autonomous decision-making</li> </ul>                                                                                     |
| <b>Google's Federated Learning</b>   | Privacy-preserving model training       | <ul style="list-style-type: none"> <li>• Models trained on-device without raw data transmission</li> <li>• Differential privacy guarantees</li> <li>• User data never leaves local device</li> </ul>     |
| <b>Microsoft's Fairlearn Toolkit</b> | Bias detection and mitigation framework | <ul style="list-style-type: none"> <li>• Comprehensive fairness metrics</li> <li>• Mitigation algorithms with tradeoff analysis</li> <li>• Integration with Azure ML for lifecycle management</li> </ul> |

3- One example for "Cognitive Requirement Engineering for the Next Generation of Ethical Artificial Intelligence Systems"?

### **Integrated Example: CRE for Next-Gen Ethical AI**

#### **Scenario: Cognitive Decision Support System for Emergency Room Triage**

##### **Cognitive Requirements:**

1. **Mental Model Alignment:** The system shall prioritize patients using cues that match experienced triage nurses' pattern recognition, weighting vital signs, chief complaints, and contextual factors in proportions validated against expert clinical judgment.
2. **Cognitive Load Management:** During high-stress periods, the interface shall reduce information density by 40% and highlight only critical decision variables, with progressive disclosure of supporting data upon request.
3. **Trust Calibration:** Confidence scores shall be presented using natural frequency formats (e.g., '3 in 10 similar patients required ICU admission') rather than percentages, with uncertainty visualized using graduated color coding validated for clinical interpretability.

##### **Ethical Integration:**

- **Fairness:** Prioritization algorithms shall be tested for demographic parity differences <5% across protected groups, with regular audits using SHAP analysis to identify feature contributions to potential disparities.
  - **Explainability:** All recommendations shall be accompanied by contrastive explanations ('Patient A prioritized over B because of X, despite Y') using language from established clinical reasoning frameworks.
  - **Accountability:** Decision logs shall capture both system recommendations and clinician overrides, with root cause analysis protocols for deviations exceeding established safety thresholds.
- 3- "Although there are many advantages to these technologies, they also bring up serious ethical issues, such as prejudice, unequal treatment, a lack of transparency, privacy dangers, and a lack of accountability" Give examples of these and justify how?

### **Ethical Issues: Examples & Mechanisms**

| Ethical Issue                | Concrete Example                                                                                           | Underlying Mechanism                                                                                             |
|------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Algorithmic Prejudice        | Amazon's recruiting tool penalizing resumes with "women's" terminology                                     | Historical bias in training data (male-dominated hiring patterns) creating discriminatory feature weights        |
| Unequal Treatment            | Healthcare algorithm allocating fewer resources to Black patients with equal medical need                  | Use of flawed proxies (healthcare costs $\neq$ health needs) that encode structural inequities                   |
| Opacity/Lack of Transparency | Credit scoring models providing only numerical scores without explanatory features                         | Trade secret protections combined with technical complexity of ensemble models creating "black box" systems      |
| Privacy Erosion              | Facial recognition databases built from scraped social media images without consent                        | Surveillance capitalism business models combined with inadequate regulatory frameworks                           |
| Accountability Deficits      | Autonomous vehicle accidents with unclear liability distribution among OEMs, software providers, and users | Legal frameworks lagging technological capabilities, coupled with complex supply chains diffusing responsibility |

4- "Because of this, moral precepts like justice, explicability, and reliability frequently stay elusive and challenging to put into practice. " Give examples of these and how?

| Moral Principle  | Implementation Challenge                                                                 | Concrete Example                                                                                                                                                                                                                                                                                                                           |
|------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Justice/Fairness | Multiple competing definitions that cannot be simultaneously satisfied                   | Criminal risk assessment algorithms must choose between: <ul style="list-style-type: none"> <li>• <b>Demographic parity:</b> Equal selection rates across groups</li> <li>• <b>Equalized odds:</b> Equal true positive/false positive rates</li> </ul> These often conflict mathematically                                                 |
| Explicability    | Different stakeholders require different explanation types and detail levels             | Medical AI system must provide: <ul style="list-style-type: none"> <li>• <b>Patients:</b> Simple causal explanations in lay terms</li> <li>• <b>Clinicians:</b> Medical rationale with evidence grading</li> <li>• <b>Regulators:</b> Technical validation of decision logic</li> </ul> Single explanation approach fails all stakeholders |
| Reliability      | Performance variability across operational environments not represented in training data | Autonomous delivery robots trained in Silicon Valley fail in Mumbai due to:                                                                                                                                                                                                                                                                |

|  |  |                                                                                                                                                                                                                       |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"> <li>• Different traffic patterns</li> <li>• Alternative pedestrian behaviors</li> <li>• Varying weather conditions</li> </ul> Edge cases become dominant cases in new environments |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

6- Problem Statement

Ethical needs including accountability, privacy, explainability, and fairness are not sufficiently captured, operationalized, or validated by current Requirement Engineering (RE) techniques for Artificial Intelligence (AI) systems. AI systems run the danger of bias, opacity, and diminished trustworthiness because ethical ideals are still abstract and lack quantifiable parameters, tool support, and lifecycle traceability.  
(Need examples for this and how)

Deficiencies in Current Requirement Engineering:

| RE Shortcoming                    | Concrete Manifestation                                                                       | Resulting Risk                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Abstract Ethical Principles       | Requirement: "System shall protect patient privacy" without specifying mechanisms or metrics | Hospital AI shares anonymized data that is re-identifiable using public records, violating HIPAA compliance |
| Lack of Quantifiable Parameters   | Requirement: "The hiring tool shall be fair" without defining fairness metrics or thresholds | Algorithm shows 20% gender disparity in selection rates, leading to discrimination lawsuits                 |
| Absence of Tool Support           | Manual ethics review processes that don't scale with agile development cycles                | Biased model reaches production because weekly sprints outpace quarterly ethics reviews                     |
| Inadequate Lifecycle Traceability | No linking between ethical principles, requirements, test cases, and code implementation     | When bias is discovered, cannot trace to specific requirement failure or identify remediation points        |